- If A, B, and C are three matrices of orders $m \times n$, $n \times p$, and $n \times p$ respectively, then $A(B + C)$ and $AB + AC$ are matrices of the same order $m \times p$ and

  $A(B + C) = AB + AC$. (Matrix multiplication is left distributive over addition)
- If A, B, and C are three matrices of orders $m \times n$, $m \times n$, and $n \times p$ respectively, then $(A + B)C$ and $AC + BC$ are matrices of the same order $m \times p$ and

  $(A + B)C = AC + BC$. (Matrix multiplication is right distributive over addition).
- If A, B are two matrices of orders $m \times n$ and $n \times p$ respectively and α is scalar, then $\alpha(AB) = A(\alpha B) = (\alpha A)B$ is a matrix of order $m \times p$.
- If I is the unit matrix, then $AI = IA = A$ (I is called multiplicative identity).

7.2.5 Operation of Transpose of a Matrix and its Properties

Definition 7.16

The **transpose** of a matrix is obtained by interchanging rows and columns of A and is denoted by A^T.

More precisely, if $A = [a_{ij}]_{m\times n}$, then $A^T = [b_{ij}]_{n\times m}$, where $b_{ij} = a_{ji}$ so that the $(i, j)^{\text{th}}$ entry of A^T is a_{ji}.

For instance,

$$A = \begin{bmatrix} 1 & \sqrt{2} & 4 \\ -8 & 0 & 0.2 \end{bmatrix} \text{ implies } A^T = \begin{bmatrix} 1 & -8 \\ \sqrt{2} & 0 \\ 4 & 0.2 \end{bmatrix}$$

We state a few basic results on transpose whose proofs are straight forward.

For any two matrices A and B of suitable orders, we have

(i) $(A^T)^T = A$ (ii) $(kA)^T = kA^T$ (where k is any scalar)

(iii) $(A+B)^T = A^T + B^T$ (iv) $(AB)^T = B^T A^T$ (reversal law on transpose)

Example 7.12

If $A = \begin{bmatrix} 4 & 6 & 2 \\ 0 & 1 & 5 \\ 0 & 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & -1 \\ 3 & -1 & 4 \\ -1 & 2 & 1 \end{bmatrix}$,

verify (i) $(AB)^T = B^T A^T$ (ii) $(A+B)^T = A^T + B^T$ (iii) $(A-B)^T = A^T - B^T$ (iv) $(3A)^T = 3A^T$

Solution

(i) $$AB = \begin{bmatrix} 4 & 6 & 2 \\ 0 & 1 & 5 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \\ 3 & -1 & 4 \\ -1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 16 & 2 & 22 \\ -2 & 9 & 9 \\ 7 & 1 & 14 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} 16 & -2 & 7 \\ 2 & 9 & 1 \\ 22 & 9 & 14 \end{bmatrix} \quad \ldots (1)$$

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